12. Hill Cipher

Program

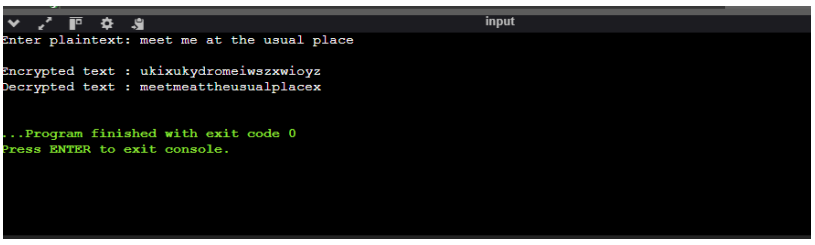
```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
int main()
{
   int key[2][2] = {{9,4},{5,7}};
   int inv[2][2] = {{5,12},{15,25}};
   char text[200], p[200], c[200], d[200];
   int i, j = 0, n;
   printf("Enter plaintext: ");
   fgets(text, sizeof(text), stdin);
   for(i=0; text[i]!='\0'; i++)
      if(isalpha(text[i]))
         p[j++] = tolower(text[i]);
   if(j % 2 != 0)
      p[j++] = 'x';
   p[j] = '\0';
   n = j;
   for(i=0; i<n; i+=2)
   {
      c[i]   = ((key[0][0]*(p[i]-'a') +
               key[0][1]*(p[i+1]-'a')) % 26) + 'a';
```

```
    c[i+1] = ((key[1][0]*(p[i]-'a') +
            key[1][1]*(p[i+1]-'a')) % 26) + 'a';
  }
  c[n] = '\0';

  for(i=0; i<n; i+=2)
  {
    d[i]  = ((inv[0][0]*(c[i]-'a') +
            inv[0][1]*(c[i+1]-'a')) % 26) + 'a';

    d[i+1] = ((inv[1][0]*(c[i]-'a') +
            inv[1][1]*(c[i+1]-'a')) % 26) + 'a';
  }
  d[n] = '\0';

  printf("\nEncrypted text : %s\n", c);
  printf("Decrypted text : %s\n", d);

  return 0;
}
```

OUTPUT